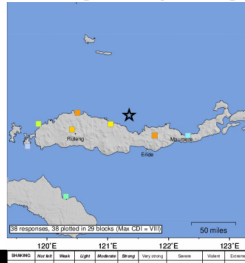


News Weekly

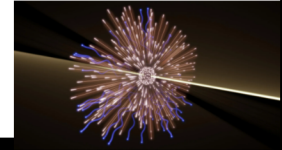
7.7 magnitude earthquake strikes Flores Island, Indonesia, killing 38 people

A magnitude-7.7 earthquake struck off Indonesia's eastern Flores region, prompting significant damage and casualties. The epicenter was located 42 miles north-northwest of Ende in East Nusa Tenggara province. Reports indicate 38 fatalities, with thousands displaced and multiple injuries. Maumere, a key town, experienced severe impacts, including fatalities and trapped residents. The quake triggered landslides, blocking roads, and damaged infrastructure, including hospitals, where patients were evacuated outdoors for safety. Yohanna Embu from Sikka regency described widespread damage and panic as locals sought higher ground. The National Disaster Management Agency initially issued a tsunami warning, though Australia reported no threat. The quake underscores Indonesia's vulnerability to seismic activity, situated along the Pacific "Ring of Fire." (NBC)



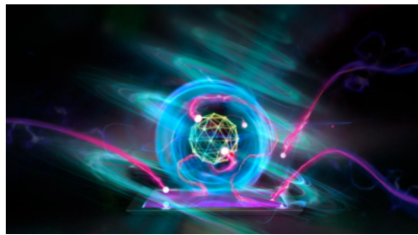
Scientists detect a nuclear reactor's ghostly afterglow for the first time

The Double Chooz collaboration has achieved a groundbreaking measurement of lingering antineutrino emissions from shut-down nuclear reactors, as published in *Physical Review Letters*. This study, led by Anthony O'Neill and Thierry Lasserre from the Max-Planck-Institut für Kernphysik, demonstrates that antineutrino detectors can provide critical insights into reactor activity even when reactors are not operational. Conducted at the Chooz nuclear power plant in France, the experiment used a detector located 400 meters from the reactors to record antineutrino interactions in a liquid scintillator. Over 17.2 days, approximately 100 antineutrino events were linked to residual radioactivity, aligning with simulations of nuclear fuel decay. The findings highlight the potential of antineutrino detectors in reactor monitoring, nuclear safety, and spent-fuel management, offering new avenues for ensuring compliance and enhancing safety protocols. (Science Daily)



World's first superconducting quantum heat engine could help unlock massive quantum computers

Researchers at Aalto University have created the first superconducting quantum heat engine, combining quantum mechanics and thermodynamics in a groundbreaking experiment. This device uses a transmon qubit, a resonator, and a quantum refrigerator to achieve cyclic operation under ultracold conditions, demonstrating the conversion of minimal heat into work. By recreating an Otto cycle within a superconducting circuit, the engine efficiently controls heat flow and produces measurable work. This innovation could significantly advance quantum computing by enabling autonomous hardware, thus reducing the complexity and cost of managing large-scale quantum systems. It aligns with Finland's goal of developing a quantum computer with one thousand logical qubits by 2035, potentially eliminating the need for costly, noise-inducing microwave cables. (Science Daily)



Construction workers uncover \$10 million in gold while building sewer under former brewery

In Belgium, construction workers uncovered \$10.4 million in gold while building a sewer beneath a former brewery in Sint-Gillis-Dendermonde. Initially mistaking the gold for 1 Euro coins, worker Kobe expressed surprise at the unexpected find. The gold, consisting of coins and bars, was placed in a high-security federal vault by local police. Under Belgian law, the rightful owner has five years to claim it. Although the workers can't keep the gold, they hope for a finder's fee. The site is secured, and the discovery could aid the welfare organization that owns the building. Police hope the treasure positively impacts lives. (Fox)



Scientists solve the mystery of a brain "switch" that can trigger weight loss in opposite ways

Scientists have made a breakthrough in understanding how two opposing methods of targeting the same brain receptor can both facilitate weight loss, potentially leading to more effective obesity treatments. A study published in *Nature Metabolism* revealed that the effects depend on the brain region being targeted. In mice, activating the receptor in the brainstem decreased appetite, while blocking it in the hypothalamus yielded similar weight loss through a different mechanism. Over a billion people worldwide live with obesity, which increases the risk for conditions such as type 2 diabetes and cardiovascular diseases. While diet and exercise are vital, substantial weight loss is often challenging. The latest weight loss medications target specific receptors like the glucagon-like peptide 1 receptor (GLP-1R) to reduce food intake and manage blood sugar. Some drugs, such as Mounjaro, activate the glucose-dependent insulinotropic polypeptide receptor (GIPR), while others, like MariTide, block it. Despite their differing mechanisms, both can aid weight loss. Cambridge researchers discovered that these GIPR-targeting drugs operate in distinct brain regions and can enhance weight loss when combined with GLP-1-based treatments. (Science Daily)



Jokes

What did the parrotsay to the clown?

“Ma-caw”.

Scoville Doodle



11					9		6		12	1	7
7		9		11	8			5	10		
	8				10	7		9	11		2
3	12		2	4		11		10		7	
		4			2	10	7			5	
	1		7						3	2	11
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	5	11			7	3	8	1		12	9
2	4	7			5		10	11			
9		1					2	6	5		4
			3		1		4	2	7	9	
4	7				11		9	3	1		12

Sambal Prawn Curry

Ingredients

- - 500g fresh prawns, peeled and deveined
- - 100g shallots, finely sliced
- - 20g garlic cloves, minced
- - 15g red chilies, deseeded and finely chopped (adjust to taste)
- - 10g ginger, grated
- - 400ml coconut milk
- - 15g mild curry powder
- - 5g turmeric powder
- - 30g tamarind paste, mixed with 50ml warm water
- - 30ml vegetable oil



1. In a mortar and pestle or food processor, combine shallots, garlic, red chilies, and ginger to form a smooth paste.
2. Heat vegetable oil in a large pan or wok over medium heat. Add the paste and sauté for 5-7 minutes until fragrant and lightly browned.
3. Stir in the curry powder and turmeric powder, cooking for another minute until aromatic.
4. Pour in the coconut milk and bring to a gentle simmer, stirring occasionally.
5. Add the tamarind paste mixture and stir well. Let it simmer for 5 minutes to allow the flavors to meld.
6. Add the prawns to the simmering sauce and cook for 3-5 minutes, or until they turn pink and are cooked through.
7. Taste and adjust seasoning if necessary. Serve immediately with steamed rice.